

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
```

```
uploaded = files.upload()
```

Choose Files polar.jpg
polar.jpg(image/jpeg) - 25894 bytes, last modified: 9/2/2026 - 100% done
 Saving polar.jpg to polar.jpg

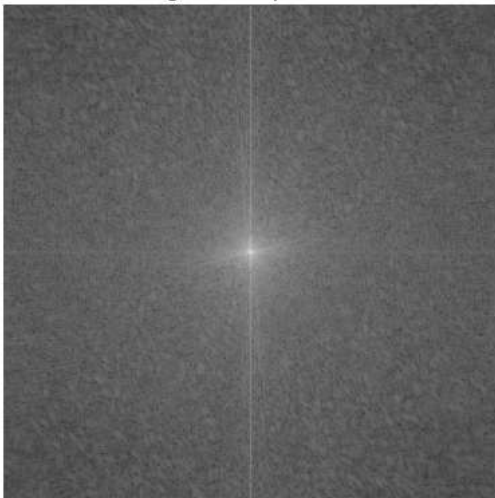
```
filename = list(uploaded.keys())[0]
img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)
plt.imshow(img, cmap='gray')
plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```

Original Grayscale Image



```
dft = np.fft.fft2(img)
dft_shift = np.fft.fftshift(dft)
magnitude_spectrum = 20 * np.log(np.abs(dft_shift) + 1)
plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("Magnitude Spectrum")
plt.axis("off")
plt.show()
```

Magnitude Spectrum



```
#low pass
new_spectrum = magnitude_spectrum
```

```

rows, cols = img.shape
crow, ccol = rows // 2, cols // 2
radius = 50
mask_lpf = np.zeros((rows, cols), np.uint8)
cv2.circle(mask_lpf, (ccol, crow), radius, 1, -1)
filtered_lpf = dft_shift * mask_lpf
img_lpf = np.fft.ifft2(np.fft.ifftshift(filtered_lpf))
img_lpf = np.abs(img_lpf)
plt.imshow(img_lpf, cmap='gray')
plt.title("Low-Pass Filtered Image")
plt.axis("off")
plt.show()

#high pass
mask_hpf = np.ones((rows, cols), np.uint8)
cv2.circle(mask_hpf, (ccol, crow), radius, 0, -1)
filtered_hpf = dft_shift * mask_hpf
img_hpf = np.fft.ifft2(np.fft.ifftshift(filtered_hpf))
img_hpf = np.abs(img_hpf)
plt.imshow(img_hpf, cmap='gray')
plt.title("High-Pass Filtered Image")
plt.axis("off")
plt.show()

```

Low-Pass Filtered Image



High-Pass Filtered Image



```

plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(magnitude_spectrum, cmap='gray')

```

```
plt.title("Magnitude Spectrum")
plt.axis("off")

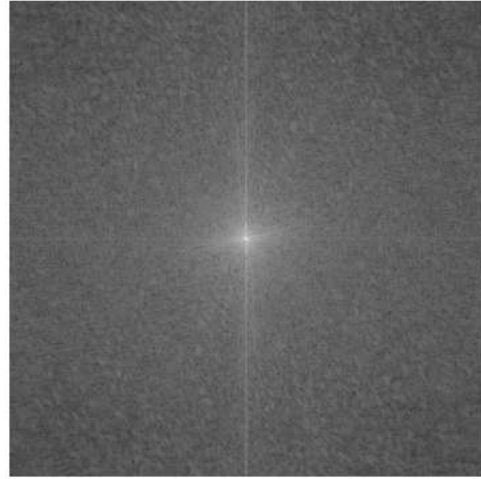
plt.subplot(2, 2, 3)
plt.imshow(img_lpf, cmap='gray')
plt.title("Low-Pass Filter")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(img_hpf, cmap='gray')
plt.title("High-Pass Filter")
plt.axis("off")
plt.tight_layout()
plt.show()
```

Original Image



Magnitude Spectrum



Low-Pass Filter



High-Pass Filter

